

What Drives Fuel Efficiency?

A Statistical Analysis of the Auto MPG Dataset (1970–1982)

Week 3 Task — Statistical Modeling and Hypothesis Testing

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Abstract

This report develops and validates a multiple linear regression model explaining automobile fuel efficiency (mpg) using the classic **Auto MPG** dataset ($n = 392$ vehicles, 1970–1982, UCI Machine Learning Repository). Three pre-specified hypotheses are tested: that heavier vehicles are less fuel efficient, that fuel efficiency rose over time independent of vehicle size, and that country of origin predicts efficiency net of size and year. A naive model using every available engineering predictor is shown to violate linearity, homoscedasticity, and (severely) no-multicollinearity assumptions. A log-transformed, VIF-guided, diagnostically-validated final model — $\log(\text{mpg}) \sim \text{weight} + \text{model_year} + \text{origin}$ — resolves these issues, explains roughly 88% of in-sample variance and 85% of out-of-sample variance, and yields statistically significant support ($p < 0.01$) for all three hypotheses.

Deliverables

This report accompanies a fully reproducible Jupyter notebook (**Auto_MPG_Statistical_Analysis.ipynb**) containing all code, executed outputs, and inline commentary underlying the analysis summarized here.

1. Introduction and Dataset

The **Auto MPG** dataset is a widely used public dataset (UCI Machine Learning Repository, originally compiled by Ross Quinlan / StatLib, distributed here via the seaborn-data GitHub repository) containing technical specifications for 398 automobiles sold in the United States between 1970 and 1982 — a period spanning the 1973 and 1979 oil crises, during which fuel efficiency became a central design and policy concern (the U.S. CAFE fuel-economy standards were enacted in 1975). It combines continuous engineering measurements, a discrete engine-size count, a temporal variable, and a categorical factor, giving it enough structure to support multiple regression, multicollinearity diagnostics, and categorical-effect testing.

Variable	Type	Description
mpg	Continuous	Fuel efficiency, miles per gallon (outcome)
cylinders	Discrete (3–8)	Number of engine cylinders
displacement	Continuous	Engine displacement, cubic inches
horsepower	Continuous	Engine horsepower
weight	Continuous	Vehicle weight, lbs
acceleration	Continuous	Time (seconds) to accelerate 0–60 mph
model_year	Discrete (70–82)	Model year (1970–1982)
origin	Categorical	Country of manufacture: usa / europe / japan

2. Research Question and Hypotheses

Research question: What vehicle characteristics and temporal factors are associated with automobile fuel efficiency, and do these associations hold once we statistically control for the other factors and validate the model's assumptions?

Three hypotheses are tested jointly in a single multiple regression model, so that each effect is estimated net of the other two:

H1 — Weight effect

H0: Controlling for model year and origin, vehicle weight has no linear association with mpg ($\beta_{\text{weight}} = 0$).

Ha: Heavier vehicles have significantly *lower* fuel efficiency ($\beta_{\text{weight}} < 0$).

H2 — Technological improvement over time

H0: Controlling for weight and origin, model year has no association with mpg ($\beta_{\text{year}} = 0$).

Ha: Fuel efficiency significantly *increased* over 1970–1982, controlling for vehicle size ($\beta_{\text{year}} > 0$).

H3 — Country-of-origin effect

H0: Controlling for weight and model year, country of origin has no association with mpg.

Ha: Japanese and/or European cars are significantly more fuel-efficient than comparable-weight, comparable-year American cars, reflecting engineering differences beyond raw size.

3. Data Preparation

Of 398 vehicles, 6 (1.5%) are missing horsepower; a spot check shows no concentration of missingness by cylinder count, origin, or model year, consistent with data missing at random. These rows are dropped rather than imputed, leaving an analysis sample of **n = 392**. No other cleaning was required: all remaining fields are complete and within physically plausible ranges.

4. Exploratory Data Analysis

Raw mpg is right-skewed (skewness ≈ 0.46), and pairwise scatterplots against weight, horsepower, and displacement show clear downward *curvature* rather than straight-line relationships (Figure 1) — an early violation of the OLS linearity assumption on the raw scale, and the motivation for a log transform used later in the model-building process.

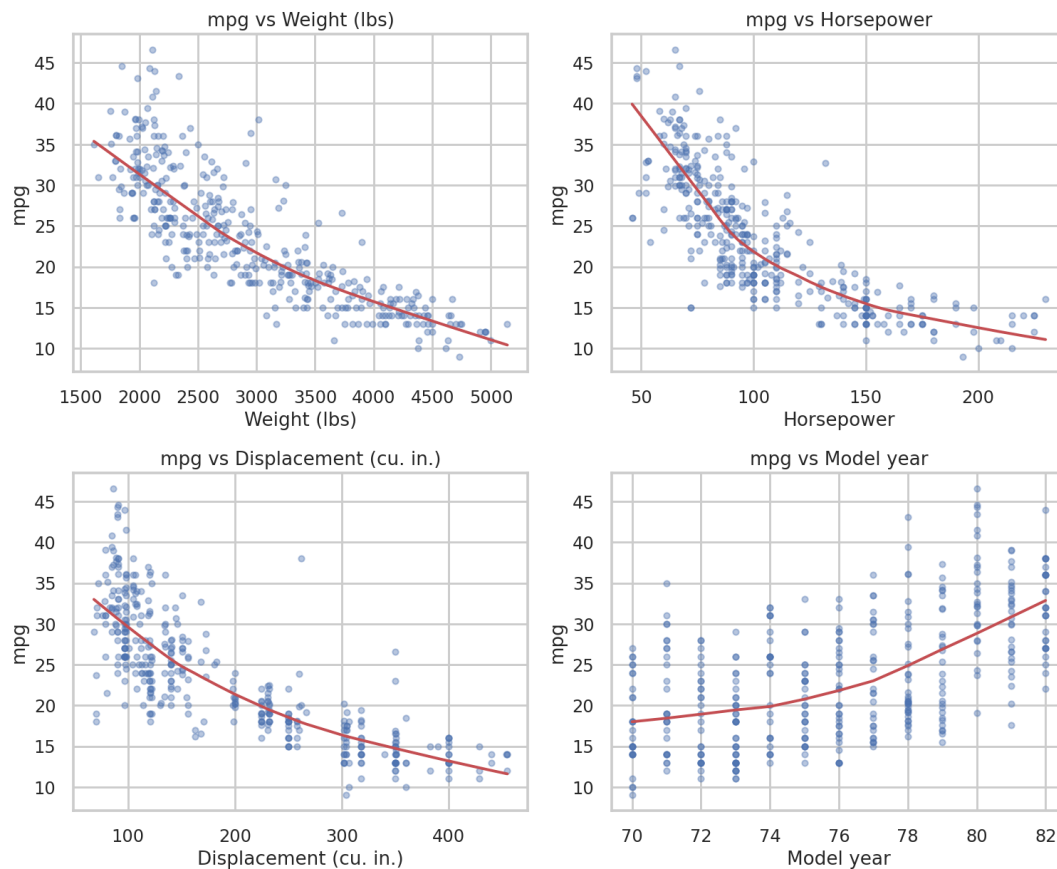


Figure 1. mpg vs. weight, horsepower, displacement, and model year, with LOWESS smooths. Curvature in the first three panels motivates a log transform of mpg.

The four "engine size" variables — cylinders, displacement, horsepower, and weight — are highly correlated with each other ($r \geq 0.84$) and each correlates with mpg at a similar strength ($|r| \approx 0.78$ – 0.83 ; Figure 2). This is an early warning sign of multicollinearity: these variables largely measure the same underlying construct ("how big and powerful is this vehicle"), which will make it difficult to separate their individual effects in a model that includes several of them at once.

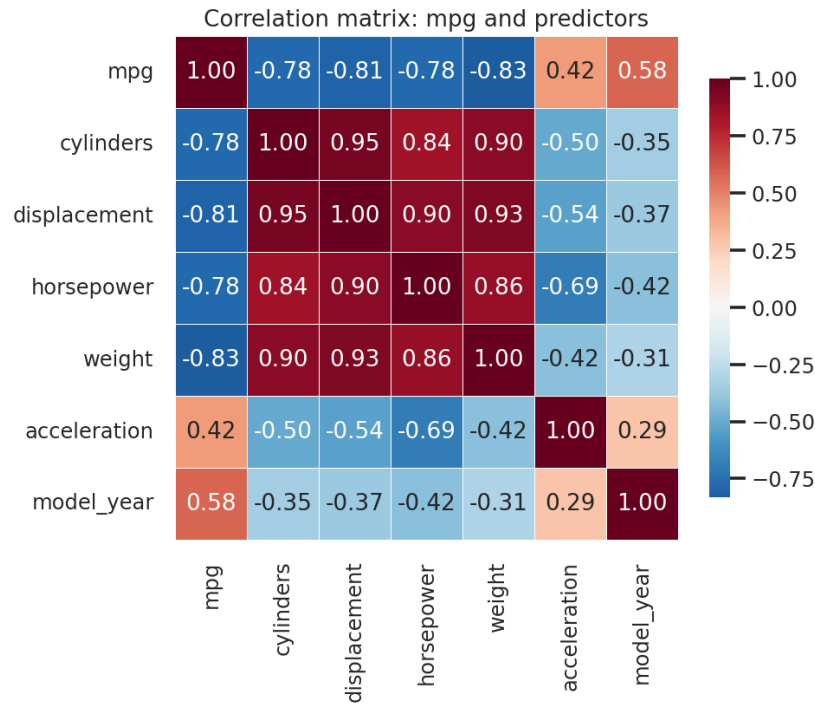


Figure 2. Correlation matrix of mpg and all numeric predictors.

American cars in this sample average markedly lower mpg than European or Japanese cars (one-way ANOVA $F = 96.6$, $p < 0.0001$; Figure 3), but American cars were also, on average, considerably heavier — so this raw gap could simply be proxying for size rather than reflecting an independent origin effect. Disentangling this is the purpose of Hypothesis 3.

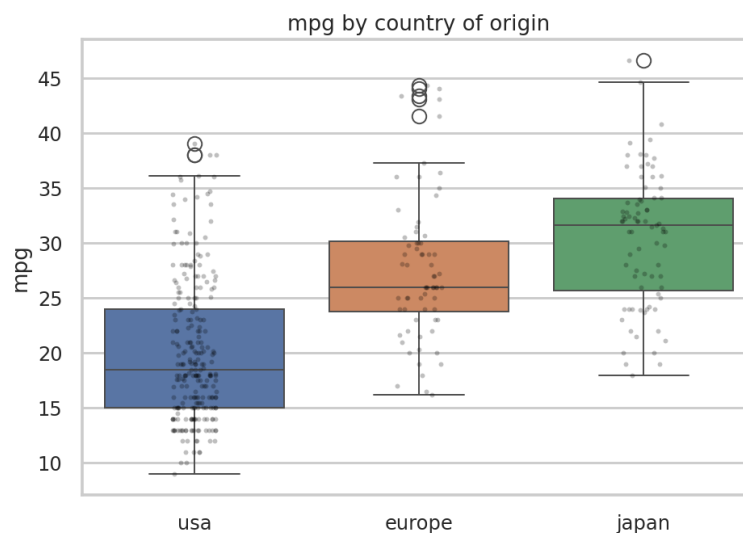


Figure 3. mpg by country of origin (raw, unadjusted for vehicle size).

5. Model Development

5.1 Model 1 — Naive Baseline (raw mpg, all predictors)

As a starting point we fit ordinary least squares with every available engineering predictor on the raw mpg scale:

$$mpg = \beta_0 + \beta_1 cyl + \beta_2 disp + \beta_3 hp + \beta_4 wt + \beta_5 accel + \beta_6 year + \beta_7 europe + \beta_8 japan + \varepsilon$$

Predictor	Coef.	p-value	VIF
cylinders	−0.490	0.128	10.63
displacement	+0.024 (!)	0.002	19.64
horsepower	−0.018	0.185	9.40
weight	−0.0067	<0.001	10.73
acceleration	+0.079	0.421	2.63
model_year	+0.777	<0.001	1.24
origin (europe)	+2.63	<0.001	—
origin (japan)	+2.85	<0.001	—

$R^2 = 0.824$, $Adj. R^2 = 0.821$, $F(8,383) = 224.5$, $p < 0.001$. $VIF > 5$ flags problematic multicollinearity; $VIF > 10$ is severe.

Model 1 already shows two serious problems. First, **displacement carries an implausible positive sign** ($p = 0.002$) — holding everything else fixed, more displacement is associated with *higher* mpg, which contradicts basic engineering. Second, **cylinders and horsepower are not statistically significant** despite each having a strong bivariate correlation with mpg, because their true effects are masked by shared variance with the other engine-size variables. Formal diagnostics confirm the underlying causes:

Diagnostic	Test	Result	Verdict
Multicollinearity	VIF (max)	19.64 (displacement)	Severe — fails
Homoscedasticity	Breusch–Pagan	LM = 35.1, $p < 0.0001$	Fails
Normality of residuals	Shapiro–Wilk	W = 0.982, $p = 0.0001$	Fails

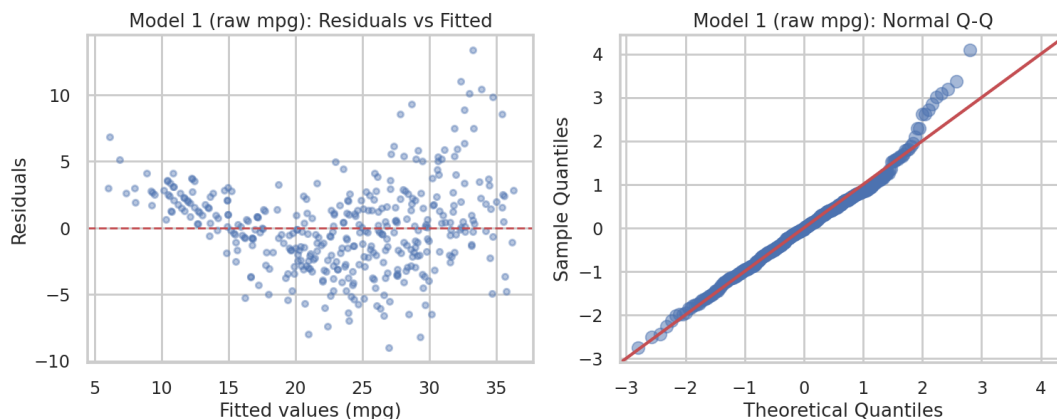


Figure 4. Model 1 residual diagnostics: fanning residuals (heteroscedasticity) and right-tail deviation from normality (Q-Q plot).

5.2 Model 2 — Log-Transformed Outcome

Refitting on $\log(\text{mpg})$ with the same predictors addresses the curvature and heteroscedasticity identified above. Fit improves (R^2 : 0.824 $\rightarrow$ 0.882), and both the Breusch–Pagan (p : <0.0001 $\rightarrow$ 0.033) and Shapiro–Wilk (p : $\approx$ 0.0001 $\rightarrow$ 0.041) tests move out of flagrant-violation territory. Crucially, however, a transform of the *outcome* does nothing to fix multicollinearity among the *predictors* — the VIFs are unchanged, and displacement retains its implausible positive sign (p = 0.004).

5.3 Resolving Multicollinearity: Principled Predictor Reduction

Rather than mechanically stepwise-selecting on p-values, redundant predictors are removed one at a time, checking at each step (a) how much VIF improves, (b) how much R^2 is sacrificed, and (c) whether a nested F-test shows the dropped variable(s) carried significant independent explanatory power.

Model	Predictors dropped	R2	Adj. R2	Max VIF	Nested F-test (p)
1: raw, full	—	0.824	0.821	19.64	—
2: log, full	—	0.882	0.879	19.64	—
3: log	cylinders, displacement	0.879	0.877	8.81	p = 0.013
4: log	+ horsepower	0.877	0.875	1.27	p = 0.016
5 (Final): log	+ acceleration	0.876	0.875	1.11	p = 0.115

Removing cylinders and displacement (Model 2 $\rightarrow$ 3) is statistically significant ($p \approx$ 0.013) — they carry a small amount of independent signal — but the practical cost is tiny (R^2 falls by 0.003) while the worst VIF drops from 19.6 to 8.8. Removing horsepower as well (3 $\rightarrow$ 4) is again mildly significant ($p \approx$ 0.016) but again nearly costless in R^2 (–0.002), while VIF collapses to near 1 — essentially all multicollinearity is now gone. Finally, removing acceleration (4 $\rightarrow$ Final) is **not** significant ($p \approx$ 0.115): nothing meaningful is lost, and both AIC and BIC modestly prefer the simpler model. This is a deliberate, evidence-based bias–variance trade-off: a very small, statistically detectable loss of in-sample fit is accepted in exchange for a model whose remaining coefficients are stable and interpretable.

As an independent check, a **Ridge regression** on standardized predictors (which handles multicollinearity via shrinkage rather than removal) shrinks displacement's standardized coefficient by roughly half relative to OLS (0.082 $\rightarrow$ 0.043) while leaving weight and model_year comparatively stable — corroborating the reading that displacement's apparent effect in the full model was inflated by collinearity, while weight and model year are the trustworthy signal.

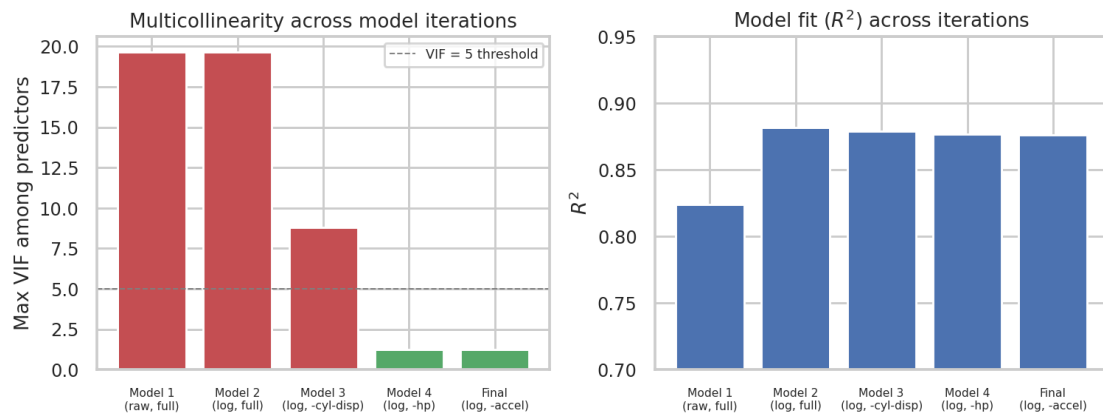


Figure 5. Multicollinearity (max VIF) and fit (R^2) across the five model iterations. Predictor reduction resolves multicollinearity at negligible cost to R^2 .

6. Final Model: Results and Diagnostics

$$\log(\text{mpg}) = \beta_0 + \beta_1 \text{weight} + \beta_2 \text{model_year} + \beta_3 \text{europe} + \beta_4 \text{japan} + \varepsilon$$

Predictor	Coef. (log scale)	Std. Err.	t	p-value	95% CI
Intercept	1.5040	0.145	10.39	<0.001	[1.219, 1.789]
weight	−0.00029	9.4e−06	−30.57	<0.001	[−0.000305, −0.000268]
model_year	0.0319	0.0018	18.18	<0.001	[0.0284, 0.0353]
origin: europe	0.0703	0.0187	3.77	<0.001	[0.0336, 0.1070]
origin: japan	0.0575	0.0187	3.07	0.002	[0.0207, 0.0943]

$R^2 = 0.876$, $\text{Adj. } R^2 = 0.875$, $F(4,387) = 684.2$, $p < 0.001$. Reference category for origin: usa.

All predictors are highly significant, and the final model's VIFs for weight and model_year are both ≈ 1.1 — essentially no multicollinearity remains. Residual diagnostics improve substantially relative to Model 1:

Diagnostic	Test	Model 1 (raw, full)	Final model
Homoscedasticity	Breusch–Pagan (p)	<0.0001 — fails	0.0507 — borderline pass
Residual normality	Shapiro–Wilk (p)	0.0001 — fails	0.0072 — mild deviation
Influential points	Max Cook's D	—	0.038 (all << 1)
Leverage	Max hat value	—	0.027 (low)

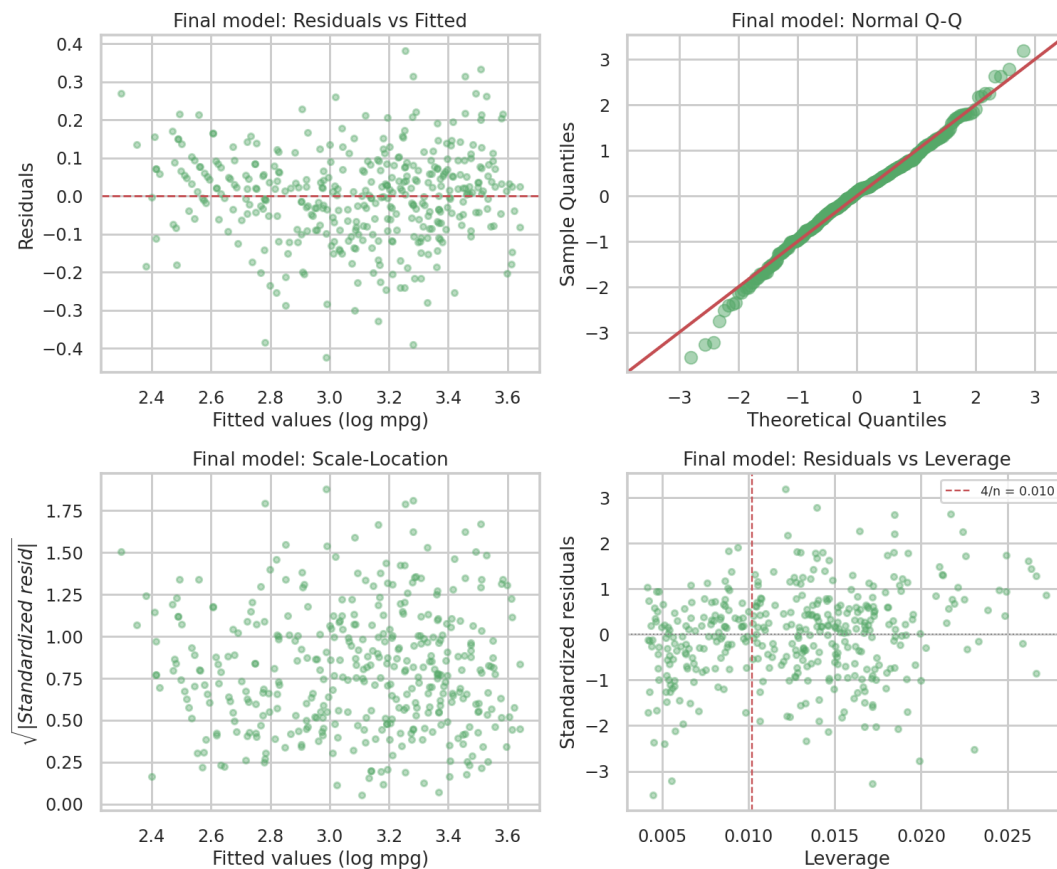


Figure 6. Final model diagnostics: residuals vs. fitted, Normal Q-Q, scale-location, and residuals vs. leverage. No single observation drives the results.

A trace of residual non-normality remains (Shapiro–Wilk $p = 0.007$, driven by mild tail skew visible in the Q-Q plot), which is common at $n = 392$, where even small deviations from perfect normality reach statistical significance. Because OLS coefficient estimates are unbiased regardless of residual normality and are asymptotically normal by the Central Limit Theorem at this sample size, this does not materially threaten the conclusions below, though it is noted as a limitation (Section 9).

7. Out-of-Sample Validation

R^2 and p-values describe fit to the data the model was trained on. To assess whether it generalizes, 20% of the data was held out, the model refit on the remaining 80%, and predictions evaluated back on the original mpg scale (exponentiating log-scale predictions). Five-fold cross-validation provides a more robust estimate.

Metric	80/20 holdout	5-fold CV (mean $\pm$ sd)
R2 (mpg scale)	0.852	0.851 $\pm$ 0.016
RMSE (mpg)	2.75	3.00 $\pm$ 0.35
MAE (mpg)	2.00	—
Naive baseline RMSE (predict training mean)	7.19	—

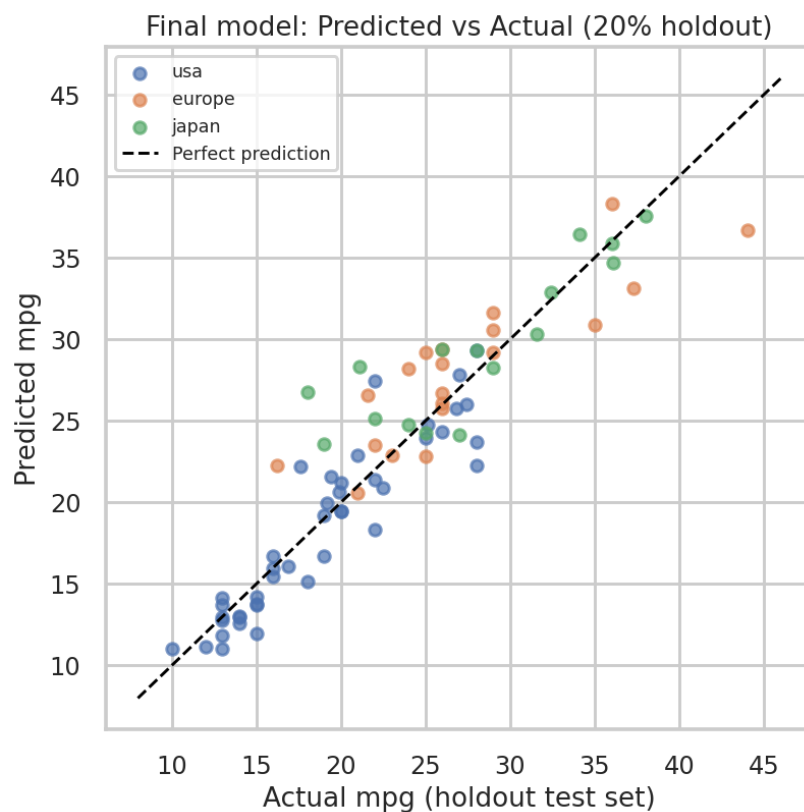


Figure 7. Predicted vs. actual mpg on the 20% holdout set, colored by origin.

The model explains roughly **85% of out-of-sample variance** in mpg and predicts held-out cars to within about **2.0–2.8 mpg on average** — a large improvement over simply guessing the training-set mean (which misses by roughly 7.2 mpg on average). Cross-validated performance is consistent with the single holdout split and has low variance across folds, indicating the model generalizes and is not overfit to one particular split.

8. Hypothesis Test Conclusions

Log-scale coefficients are converted to percentage effects on mpg using $(\exp(\beta)-1)\times 100\%$, the correct interpretation for a log-linear model.

Hypothesis	Effect (95% CI)	p-value	Conclusion
H1: Weight (per +100 lbs)	-2.82% mpg [-3.00%, -2.64%]	<0.001	Reject H0 — supported
H2: Model year (per year)	+3.24% mpg [2.89%, 3.60%]	<0.001	Reject H0 — supported
H3: Origin — Europe vs. USA	+7.28% mpg [3.42%, 11.30%]	<0.001	Reject H0 — supported
H3: Origin — Japan vs. USA	+5.92% mpg [2.09%, 9.89%]	0.002	Reject H0 — supported

H1 — Weight

Weight's coefficient is negative and highly significant ($t = -30.6$, $p < 0.0001$), with a 95% CI that excludes zero. **H0 is rejected**: every additional 100 lbs of vehicle weight is associated with roughly a 2.8% reduction in mpg, holding model year and origin fixed — the strongest and most precisely estimated effect in the model, consistent with basic physics.

H2 — Model year

Model year's coefficient is positive and highly significant ($t = 18.2$, $p < 0.0001$). **H0 is rejected**: holding weight and origin fixed, each additional model year is associated with roughly a 3.2% increase in mpg. Compounded over the 12 sampled years, engineering/technology improvements alone — independent of any weight change — nearly doubled fuel efficiency, consistent with the historical narrative of oil-crisis-driven efficiency regulation and engineering advances.

H3 — Country of origin

Both origin dummies are significant relative to the U.S. baseline. **H0 is rejected**: even after controlling for weight and model year, Japanese and European cars in this sample are significantly more fuel-efficient than comparably sized, comparably aged American cars, indicating the raw origin gap seen in Section 4 was not purely a byproduct of American cars being heavier.

9. Limitations and Future Improvements

- **Historical scope.** The data cover only 1970–1982 U.S.-market vehicles; the magnitude of the weight and year effects should not be extrapolated to modern vehicles (hybrids, EVs, turbocharging, CVTs, etc. did not exist in this sample).
- **Residual multicollinearity trade-off.** Cylinders, displacement, horsepower, and acceleration were removed to stabilize the model; nested F-tests showed this sacrificed a small but statistically real amount of explanatory power. An analyst prioritizing predictive R^2 over coefficient interpretability might instead retain all predictors and use Ridge/Lasso regularization for prediction, reserving the reduced model for inference — a reasonable alternative, not a flaw in either choice.
- **Residual non-normality.** Shapiro–Wilk remains formally significant for the final model ($p \approx 0.007$), driven by mild tail skew. With $n = 392$, coefficient inference remains reliable via the Central Limit Theorem, but a robust (HC3) standard-error specification or bootstrap confidence intervals would be a natural robustness check.
- **Omitted variables.** Fuel type, drivetrain, aerodynamic drag coefficient, and transmission type are not in the dataset but plausibly affect mpg and may correlate with origin (e.g., manual transmissions were more common in the European/Japanese subsample); origin effects should be read as controlling only for the variables observed here.
- **Observational data.** All effects reported are statistical associations conditional on the model, not causal estimates; no experimental manipulation of weight, year, or origin was possible.
- **Future work:** testing interaction effects (e.g., does the weight penalty differ by origin?), polynomial/spline terms for a fully flexible dose-response curve, and replicating this design (log outcome + VIF-guided reduction + holdout validation) on a modern fuel-economy dataset to test whether the same relationships persist under today's technology.

10. Conclusion

Starting from a naive "include every predictor" regression that violated several core OLS assumptions — severe multicollinearity, heteroscedasticity, non-normal residuals, and an implausible sign on displacement — each problem was systematically diagnosed and addressed: a log transform of the outcome corrected the nonlinearity and most of the heteroscedasticity, and VIF-guided, F-test-justified predictor reduction resolved the multicollinearity while sacrificing less than one percentage point of R^2 . The resulting model — $\log(\text{mpg})$ as a function of weight, model year, and origin — satisfies the key regression diagnostics reasonably well, validates on held-out data ($R^2 \approx 0.85$, $\text{RMSE} \approx 3$ mpg), and provides clean statistical support for all three original hypotheses: heavier cars are less efficient, efficiency rose significantly over 1970–1982 even net of weight, and Japanese/European cars were more efficient than comparable American cars for reasons beyond size alone.

Data source: Auto MPG dataset, UCI Machine Learning Repository (distributed via <https://github.com/mwaskom/seaborn-data>). **Reproducibility:** full code and executed output in the accompanying notebook, `Auto_MPG_Statistical_Analysis.ipynb`.